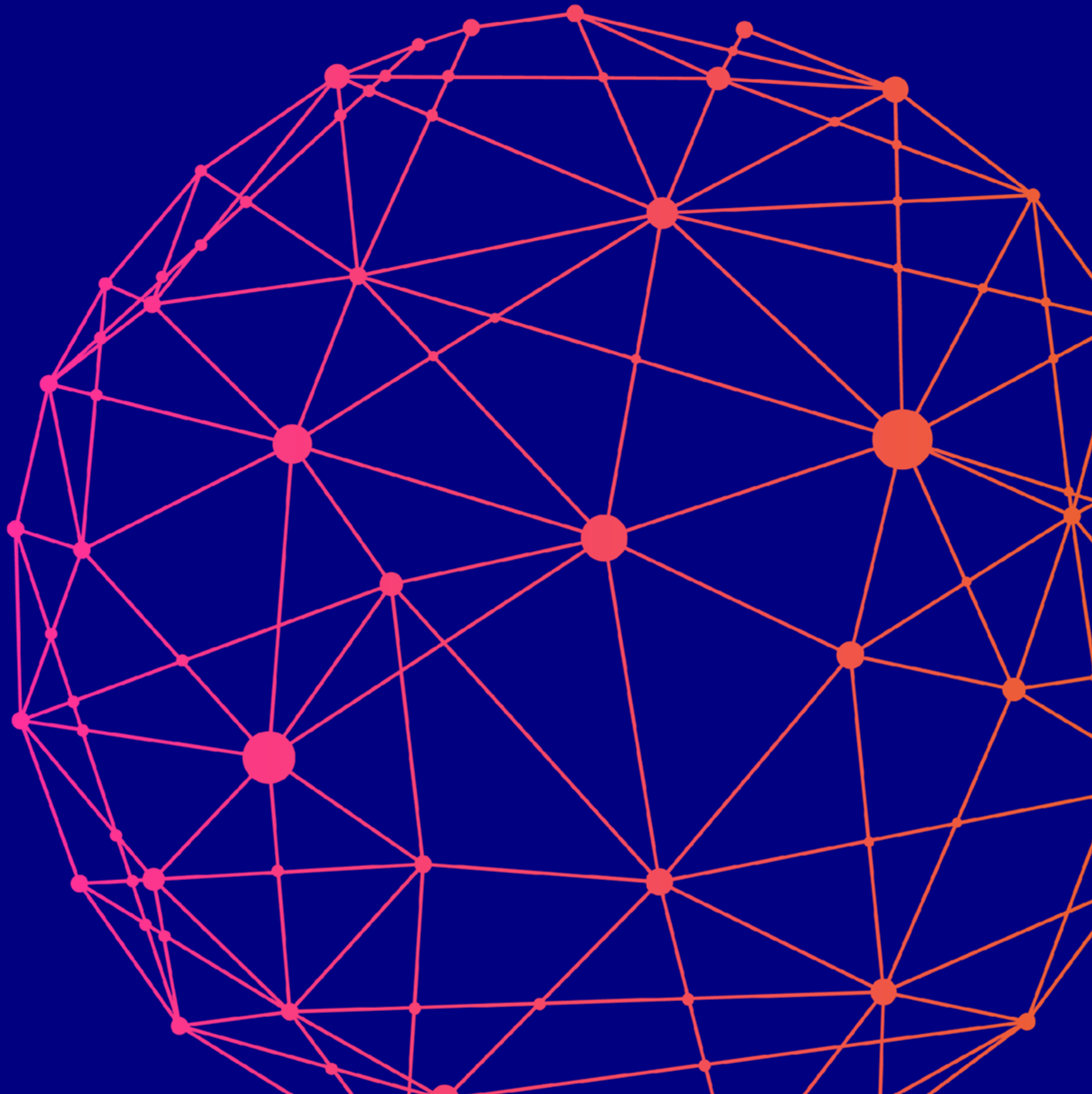
Data Spaces Symposium

Accelerating adoption. Increasing impact.

How the Berlin Declaration can push data spaces as indispensable element of digital ecosystems and AI



Ernst Stoeckl-Pukall

Data Spaces Symposium

Berlin Declaration of Friends of Industrie 2025

“Economic security is an essential precondition for digital sovereignty: only a coordinated approach between industrial, digital, and security policies can ensure Europe’s stable and sustainable leadership over time. Today, European solutions are often of smaller scale and only cover parts of the digital value chain, whereas some international competitors offer a full stack of solutions. It should be our goal that European industry has the choice to work with European technologies across the whole value chain to increase resilience as well as digital and technological sovereignty.

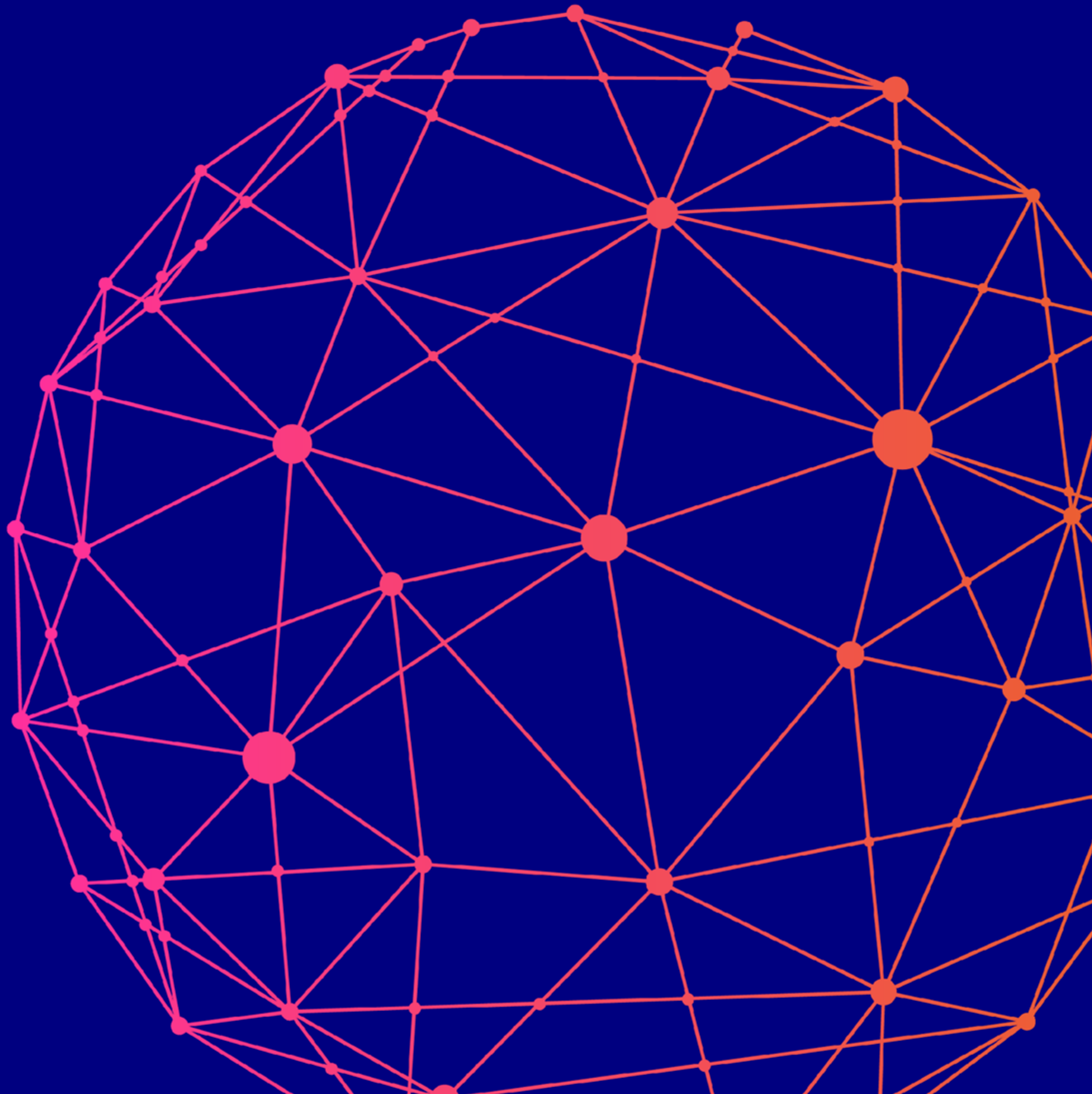
One way to address this issue efficiently is by combining existing European solutions through increased interoperability. This is not only the rationale of the IPCEI Next Generation Cloud Infrastructure and Services (IPCEI-CIS), but also of the IPCEI on Artificial Intelligence (IPCEI-AI) and for Edge Cloud Infrastructure (IPCEI-CIC) that are being discussed. Furthermore, the development of interoperable and trusted data ecosystems in Europe is key to jointly use high quality and sensitive data across companies, e.g. for innovative services and AI applications.”

Overview

What achievements?

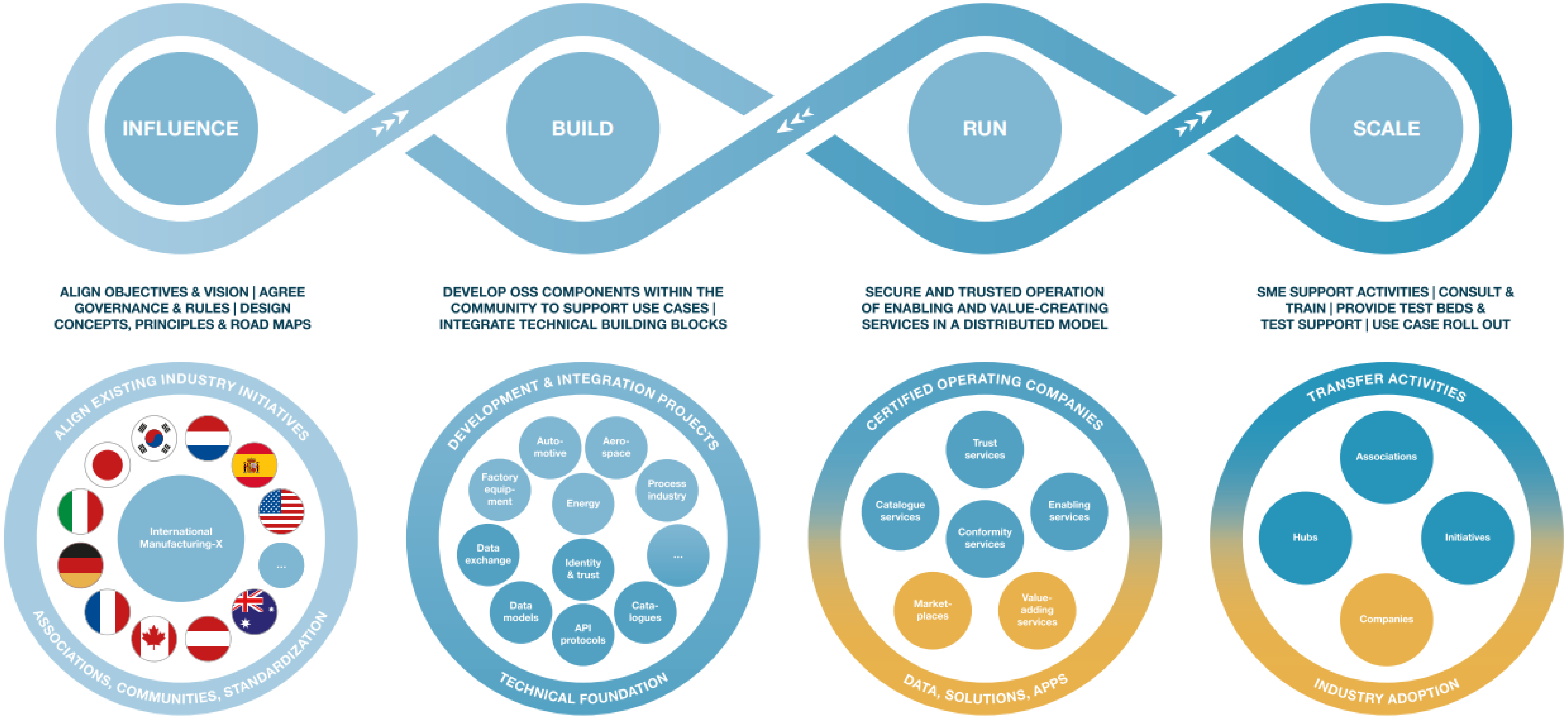
Challenges ahead

Industrial AI – Push forward?



Systemic approach

Way to organize (industrial) data ecosystems



Alignment of objectives & vision

Influence

Achievements:

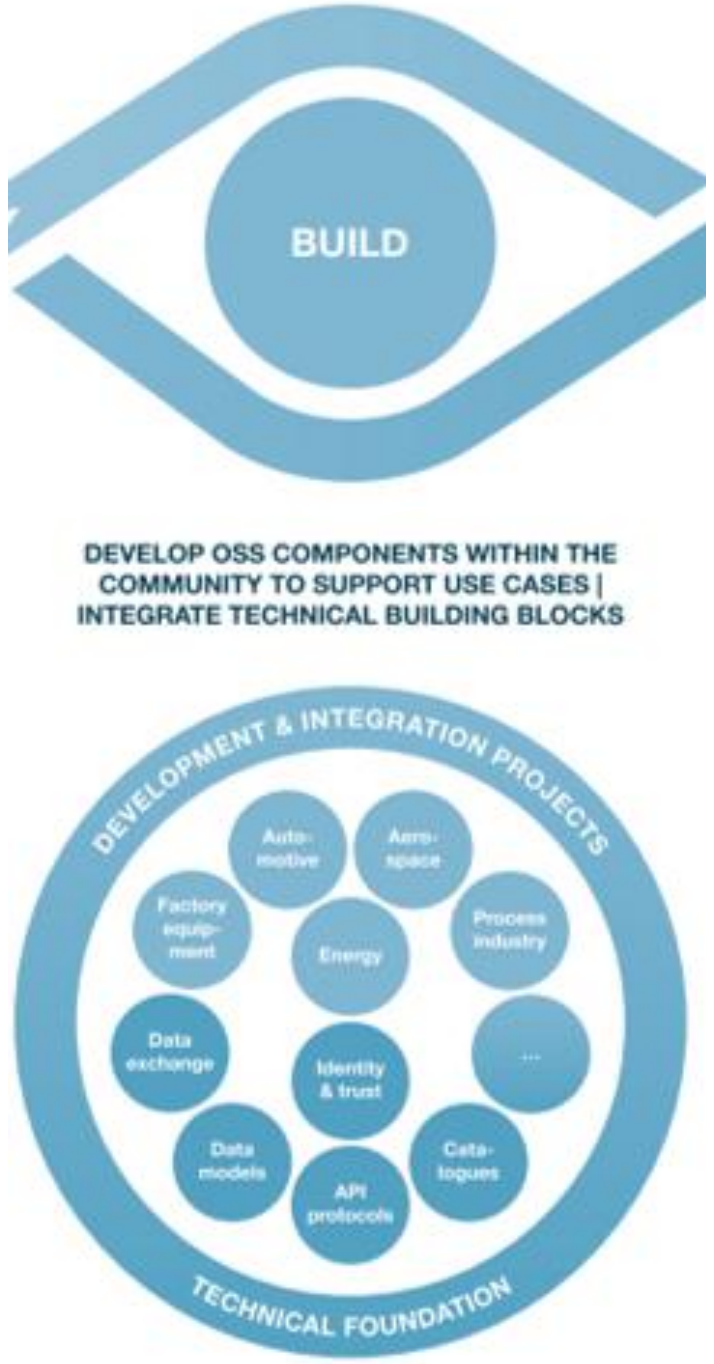
Legal entities:	Catena-X e.V., Decade-X etc
Other organizations:	e.g. IOFDS (JAPAN)
International:	International Manufacturing-X Council
Projects:	Huge number of funded projects across countries and industrial domains (aerospace, chemical, construction, etc.)

Challenges/Targets:

- Readiness to invest in organizations low/strong mandates from stakeholders required
- Systemic approach and cross domain interoperability on a global level
- Different state of maturity of projects

Integration of technical building blocks

Build



Achievements:

- Many building blocks developed: Dataspace Protocols, Trust Frameworks,…
- Eclipse OSS projects (tractus-X, XFSC…)
- Simpl

Challenges:

- Adaption of existing Code across initiatives
- Interoperability of existing solutions
- Implementation of tractus-X as possible concept for use case interoperability

Targets:

- Strong, global community processes for OSS development
- Comprehensive documentation: use case documentation, templates, source (KIT)
- Standardization (protocols)
- Overarching Reference Architecture Model defining clear interfaces
- Strategy for domain/geo specific extensions and configurations

Operation of value-creating services

Run

Achievements

First Operating Companies running!

Example: Cofinity-X (founded in the context of Catena-X, open for all dataspaces!)

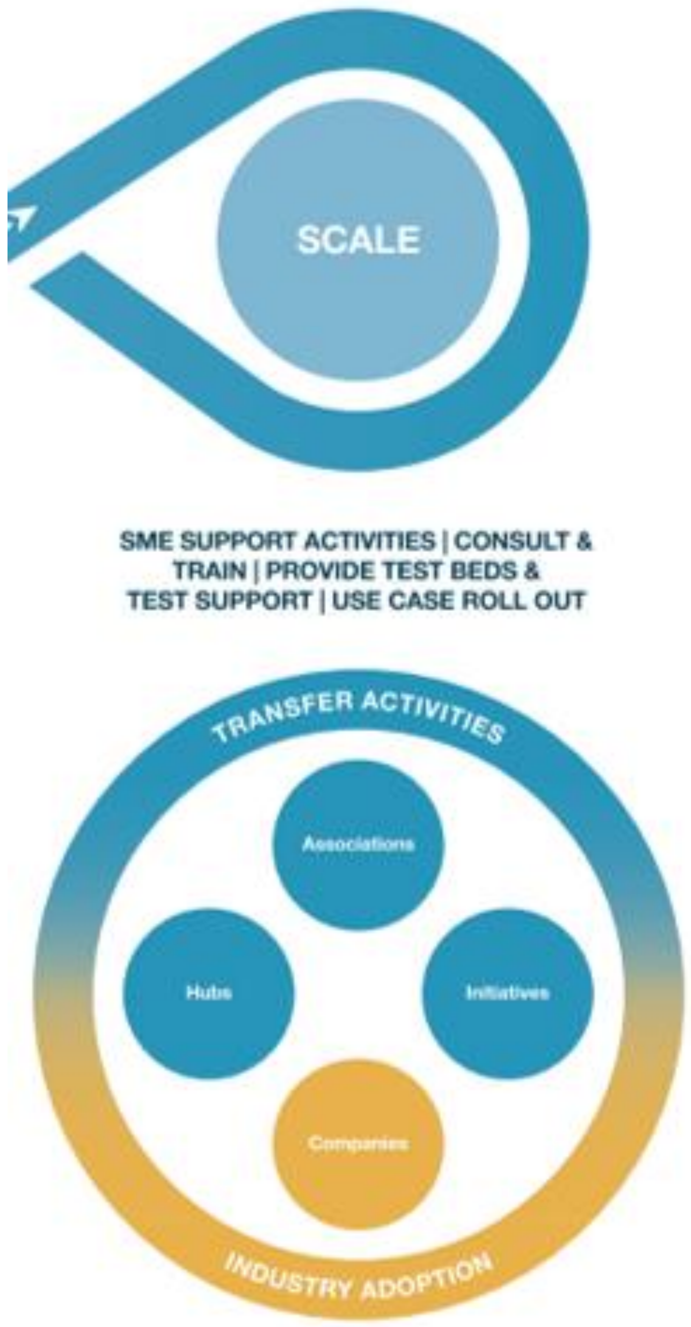
Learnings: Huge investment necessary to deliver SLA!
OpCo's necessary in advancing data ecosystems

Challenges/Targets

- Establishing a sufficient number of OpCo's
- Interoperability of OpCo's globally
- Apart from OpCo's: Competitive Services/App-offerings in Ecosystems

Roll-out and transfer (to SME's)

Scale



Achievements

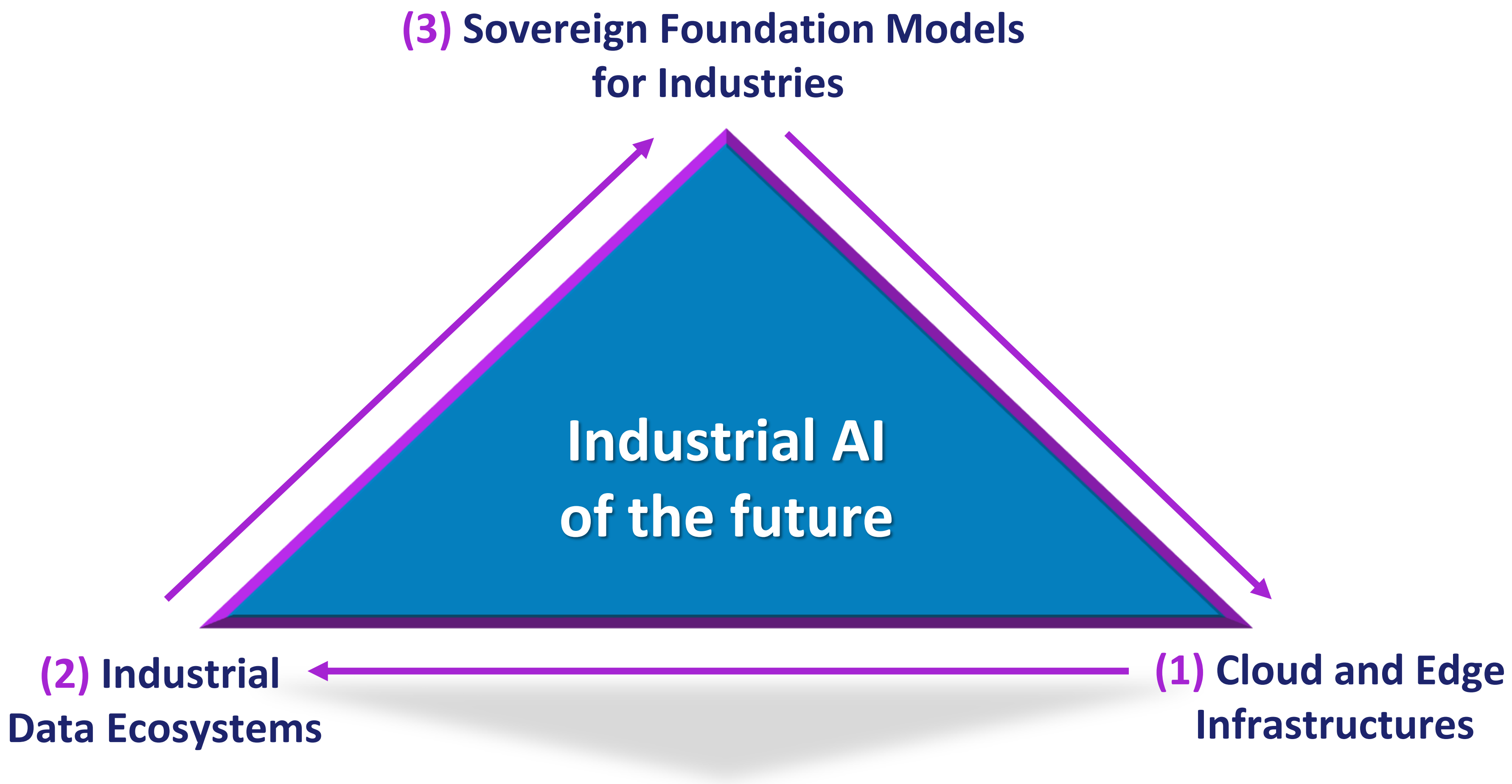
Substantial number of activities on national level; measures funded by EU-Commission etc.

Challenges

- Data readiness of companies
- Benefit of data ecosystems still not clear enough to trigger investment
- Robust business solutions not available (with exceptions)

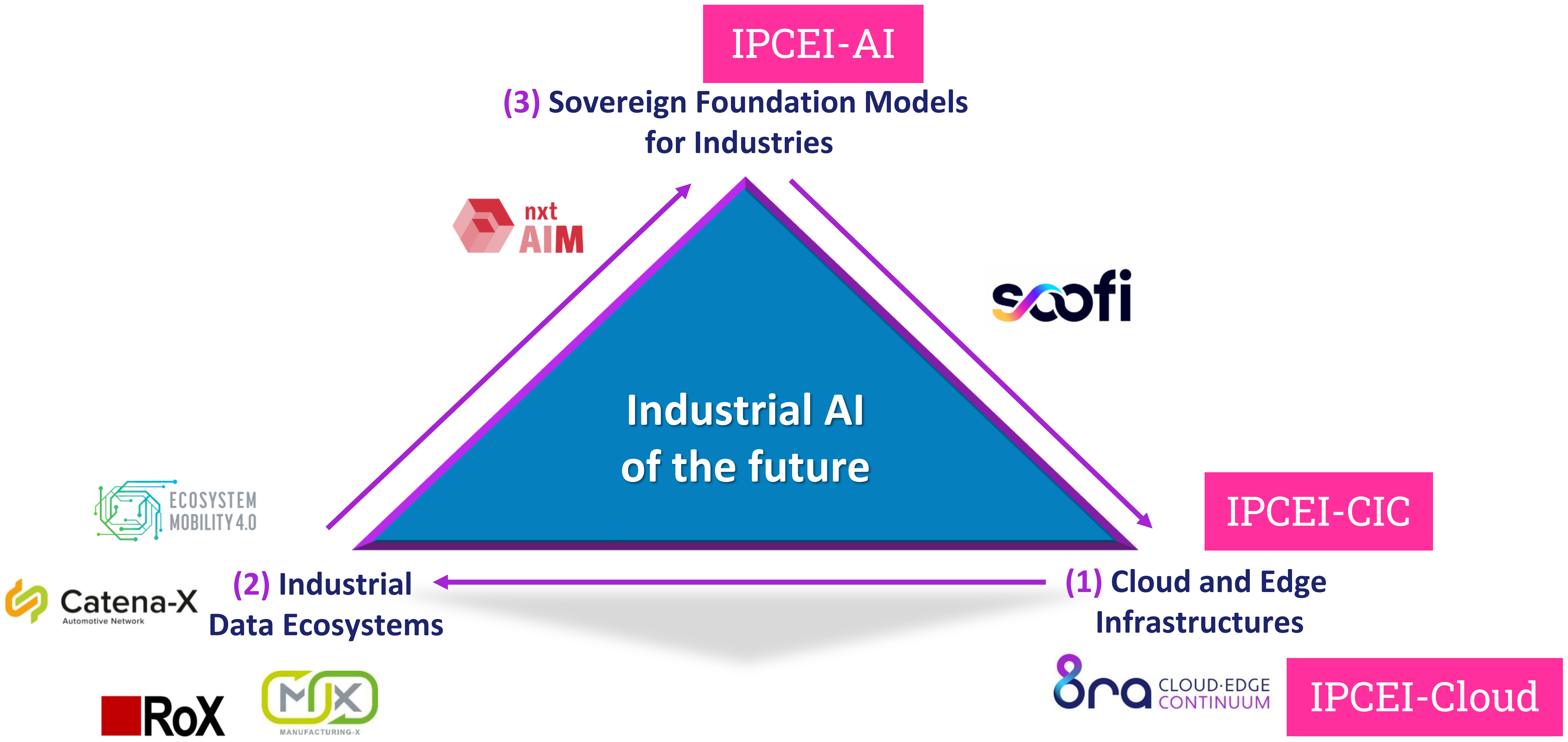
Industrial AI

Building blocks for digital sovereign industries



Industrial AI

Building blocks for digital sovereign industries



We need to act fast and together

Conclusion

1. Objectives: innovation, competitiveness and speed
2. Industrial AI is a gamechanger
3. Data ecosystems key to industrial AI – sensitive data as basis for quality models
4. Urgency to get organized – orchestration, management and operation of digital infrastructure (8ra-Initiative!), data ecosystems and industrial AI ecosystem

➢ **IPCEI AI** is the opportunity to realize the concept of distributed, trusted and federated systems for a thriving industry